

KINETARIUM: Interactive Multiplayer Games for Fulldome Projections

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ABSTRACT KINETARIUM is a new platform for interactive, collaborative fulldome shows for hundreds of people. It enables visitors to intervene spontaneously and in real time in what is happening on the dome, by using their own smartphones. Kinetarium introduces interactivity and gamification to the domes, with the visitors becoming fully immersed in the projection, just as if they were right in the middle of things. Everyone in the audience can participate in the show. Together, the players can go on missions, solve puzzles and discover new worlds – or simply try to crack the high score. In addition to dealing with scientific phenomena, the players experience how difficult the simplest tasks can be when joint decision-making, coordination, team work or compromises are required. That way, the games also teach learning processes about group dynamics or social behavior. The planetariums can thus enrich scientific content with playful and group-dynamic elements and make their program more attractive for a young, gaming and 3D savvy audience.

KEYWORDS Interactive experience; fulldome projection; local mass multiplayer; realtime rendering; immersive; 360° media; serious gaming; collaborative learning; projection mapping.

I. INTRODUCTION – OUR SOLUTION

INSTEAD of being asked to turn their phone off, our audience is encouraged to bring a fully charged phone to the show. Visitors open a website on their smartphone which serves as a controller and enables real time interaction (a large audience is shown in Fig. 1).

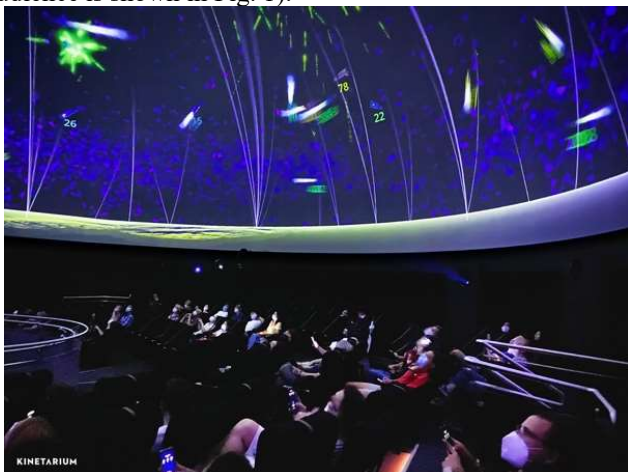


Figure 1. The star dome goes interactive: Kinetarium invites hundreds of players to interact with each other in a physical space

A web server collects all user input and forwards it to the game engine, which updates all game graphics and prepares the new image for projection.

A. LOW BARRIER OF ENTRY FOR THE PLANETARIUM VISITORS

We want to appeal to all planetarium visitors. Therefore, we have taken care not to exclude anyone by assuming gaming experience or technical understanding.

Every visitor uses their own smartphone as an input device, that way they are already familiar with the hardware. The controller is a website, not an app. There is no need to come prepared, no need to install anything. The website is loaded up quickly and can be used right away. No registration required. Any web browser can be used. The controller interface automatically changes to reflect the currently selected game or show (interface example in Fig. 2). Any buttons that aren't needed are hidden. We designed each individual controller to be usable without looking at the screen, since the action happens in the dome.

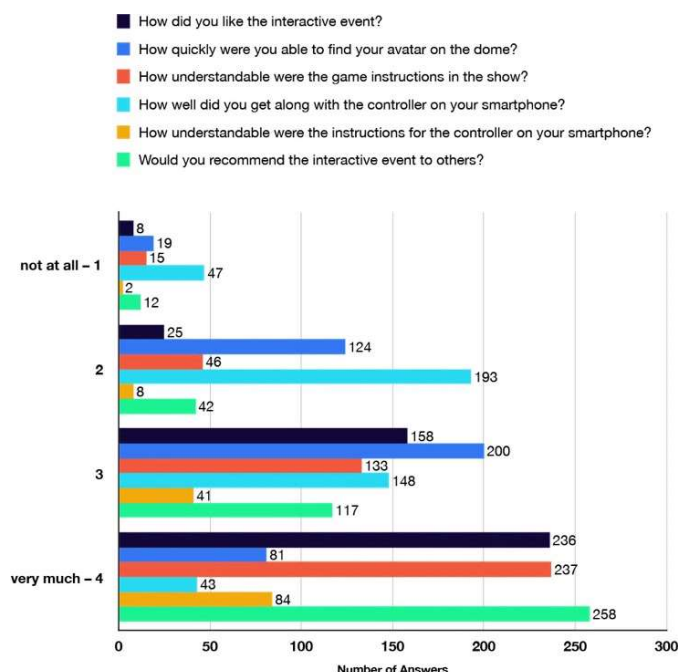


Figure 16. Feedback for the 6 different questions (in total numbers per answer)

A. INTEREST

Regarding the evaluation, it can be said that the general interest of visitors in interactive content is high. 52% of all respondents said that they liked the event very much (4 on the scale) and 56% recommended it very highly. 35 % liked the event and 26% would recommend it (3 on the scale). We could observe on site that especially after a learning phase of about 20 minutes, many visitors were gripped by ambition. In our events with an earlier prototype, however, some found the information content too low. The key here is to find the right balance for each audience. Since the interactive phases of the event take up quite a bit of time, the informative, didactic phases quickly come up short.

B. CONTROLS

Getting along with the three-dimensionality of the controls still seems to be difficult. The two-dimensional control that the joystick offers is very limited in the three-dimensional space that the dome projection represents. Here, 42% of viewers said they couldn't get along well with the controls (2 on the scale) and 10% not well at all. In the same context, a total of 31% of the players said they could not find their avatar quickly at the dome or not quickly at all (1 and 2 on the scale). According to our observations during the shows, we noticed that especially the younger visitors would usually find their way around the dome very quickly. A few visitors had difficulties with the connection or the controls. But after a short period of getting used to the system, almost all visitors were able to play along.

VIII. CHALLENGES

A. SHOWS AND GAMES

We try to strike a balance between gaming and education. The correct mix for each event can differ drastically, depending on the venue's philosophy or the current audience. Planetarium visitors have drastically different expectations, education and

experiences. Most likely, Kinetarium needs to offer a broad spectrum of events.

Cooperation is at the heart of what Kinetarium is. To enable hundreds of people to cooperate is a game design challenge that we will keep exploring. These mechanics need to work for small groups, just as well as for large ones.

B. REPLAYABILITY

Video game players are already familiar with many of the concepts of the Kinetarium, so controls and game principles hardly need to be explained. However, one major motivating factor is missing: that of personal progress. Each individual player improves within an event, but can take little of it home in the long run. The only thing left to do is to visit another event, which may be weeks in the future or may not happen at all.

So far, Kinetarium has been designed and developed as a local application for several hundred players under a common dome; a logical next step is the networking of several domes worldwide.

C. INTERACTION

Kinetarium games should be as accessible as possible. We need to make sure everyone can quickly pick up on how to open the controller website and how to use it. This will always be a priority.

Another game design challenge is that of each individual player's impact: for any game to be fun, every player needs to have the feeling that their decisions matter. To promote cooperation however, collective decisions must cancel out individual ones. This also applies to graphics: every player needs to see some kind of impact, but the collective actions are the ones that need to be highlighted.

D. TECHNICAL CHALLENGES

Planetariums are organized in very diverse ways: as museums, as private companies, as educational institutions, as non-profit organizations. This results in very different financial possibilities and requirements on the show program.

An interactive planetarium program should always have the lowest possible technical and operational requirements so that it can function in most planetariums. Usability of the system must be simple enough that it can be performed by anyone without training.

When preparing Kinetarium for a global market, we have to consider controller input latency. To keep latency within a reasonable range, multiple web servers will have to be set up around the globe.

E. COMMUNICATION FOR A NEW TYPE OF EVENT

We discovered that it is difficult to promote this new event format. There are several challenges here: on the one hand, potential visitors need to understand what they can do in an event in the first place. A classic movie poster, for example, works by showing the content as the topic of the poster. People don't need an explanation on how a movie theater works.

Since planning the first event, several forms of media have been used to promote Kinetarium events and introduce the project. We have a dedicated subpage on the planetarium's Stuttgart website, and the events can also be found in the printed program booklet including a short description. In the city of Stuttgart, we were also able to advertise some events

with various posters. Social media channels have been maintained on Instagram, Twitter and Facebook. Various postings give an overview of what the Kinetarium is and how the interaction works. Video footage of events and illustrations are primarily used for this purpose. In addition, photos and videos taken while testing in the dome show a behind-the-scenes look.

In our feedback form mentioned earlier we tried to evaluate how visitors became aware of the event as shown in Fig. 17. Online advertising and classic advertising with posters and booklets seem to have had a similar impact. This can be attributed to the diverse age and media consumption habits of planetarium visitors.

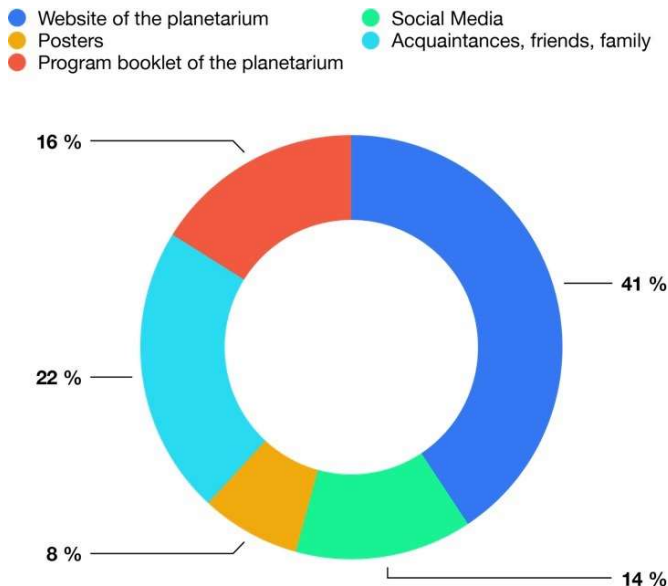


Figure 17. How visitors became aware of a Kinetarium event (in percentage)

IX. VISIONS AND PERSPECTIVES

Once these challenges are solved, a wide range of possible applications opens up, e.g. league games within a worldwide community, Escape Room applications, etc. The development of globally functioning interaction principles can in turn generate novel game ideas and application scenarios.

Another consideration for the further development of Kinetarium concerns the use of a streaming service e.g. Google or Amazon. Instead of a downloadable Kinetarium client that runs locally, the fulldome image could be rendered in the cloud and streamed to the venue. Providing the image for the dome projection via streaming would bring a few significant advantages: even less setup required by the venue, virtually no hardware requirements, and the ability to expand and update the content offering at any time.

The widespread use of high-performance smartphones with mobile internet connection enables visitors to interact easily and spontaneously via a familiar device. Availability of the mobile internet at even higher transmission rates enables live interactions without latency, which will get significantly better with the 5G standard. Another possibility to further increase the use cases of our technology, is to offer Kinetarium for non dome-shaped projections (Fig. 18). Anywhere there is an audience and a projection, Kinetarium can offer spontaneous and engaging interaction to everyone in the room.

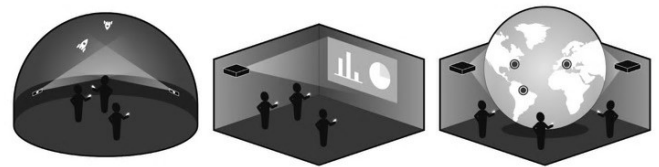


Figure 18. Dome, wall, sphere – possible projections surfaces for the Kinetarium

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Our show “In Orbit” was developed in close collaboration with the Planetarium Stuttgart as well as the German Aerospace Center Stuttgart (DLR).

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The Show “In Orbit – an interactive journey” has been awarded with the SILBERSALZ Science & Media Award in the category IMMERSIVE SCIENCE and the Horsetooth International Film Festival Award in the category Another Dimension.

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